

8

C

$$(\uparrow): N \sin \theta = mg$$

$$(\rightarrow): N \cos \theta = F_c = ma_c = m(0.8g)$$

$$\tan \theta = 1.25$$

$$\theta = 51.3^\circ$$

Option A: shift cos (0.8)

Option B: swap the angles in resolving

Option D: shift sin (0.8)

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D

The rotation of the Earth results in the acceleration of free fall being smaller than the gravitational